

Thermodynamics

Instructor

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Thermodynamics: An Engineering Approach, 6th Edition
Yunus A. Cengel, Michael A. Boles
McGraw-Hill, 2008

Chapter 8

EXERGY: A MEASURE OF WORK POTENTIAL

Mehmet Kanoglu

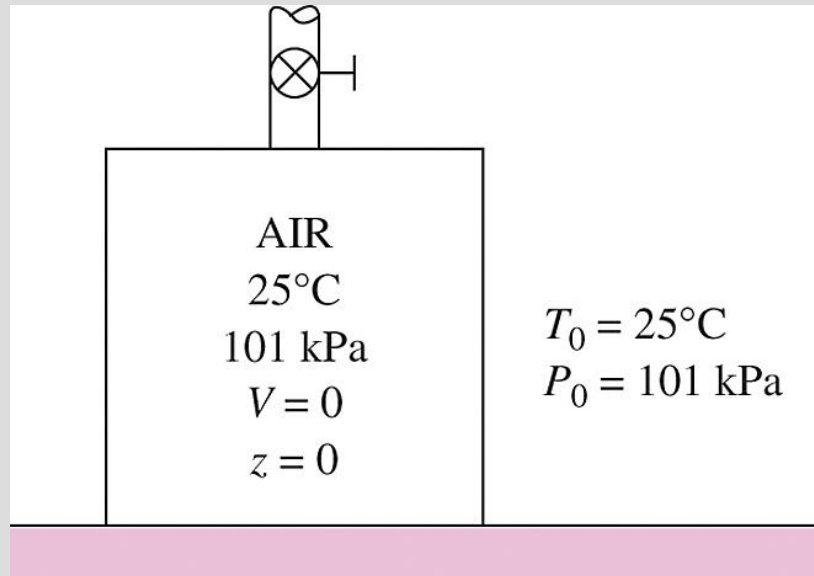
Objectives

- Examine the performance of engineering devices in light of the second law of thermodynamics.
- Define *exergy*, which is the maximum useful work that could be obtained from the system at a given state in a specified environment.
- Define *reversible work*, which is the maximum useful work that can be obtained as a system undergoes a process between two specified states.
- Define the exergy destruction, which is the wasted work potential during a process as a result of irreversibilities.
- Define the *second-law efficiency*.
- Develop the exergy balance relation.
- Apply exergy balance to closed systems and control volumes.

EXERGY: WORK POTENTIAL OF ENERGY

The useful work potential of a given amount of energy at some specified state is called *exergy*, which is also called the *availability* or *available energy*.

A system is said to be in the *dead state* when it is in thermodynamic equilibrium with the environment it is in.



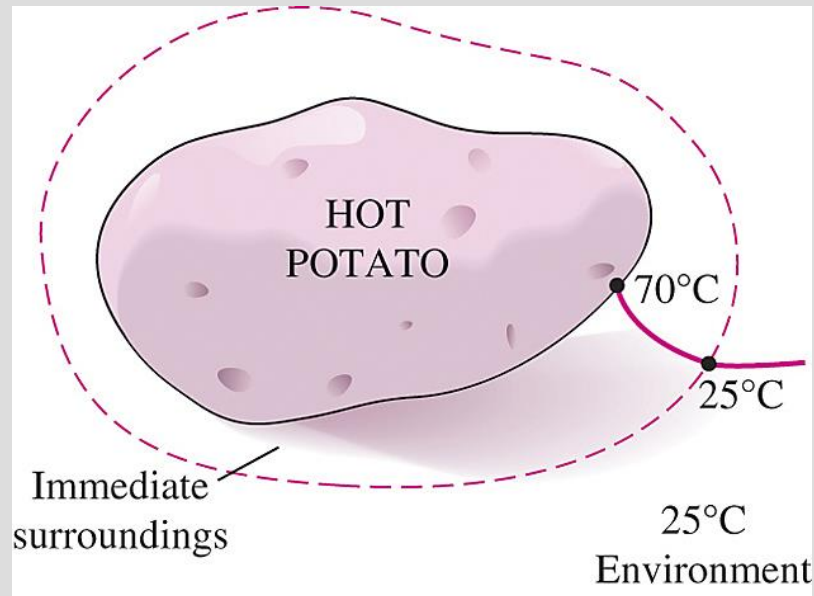
At the dead state, the useful work potential (exergy) of a system is zero.

A system that is in equilibrium with its environment is said to be at the dead state.

A system delivers the maximum possible work as it undergoes a reversible process from the specified initial state to the state of its environment, that is, the dead state.

This represents the *useful work potential* of the system at the specified state and is called exergy.

Exergy represents the upper limit on the amount of work a device can deliver without violating any thermodynamic laws.



The immediate surroundings of a hot potato are simply the temperature gradient zone of the air next to the potato.



The atmosphere contains a tremendous amount of energy, but no exergy.

Exergy (Work Potential) Associated with Kinetic and Potential Energy

Exergy of kinetic energy:

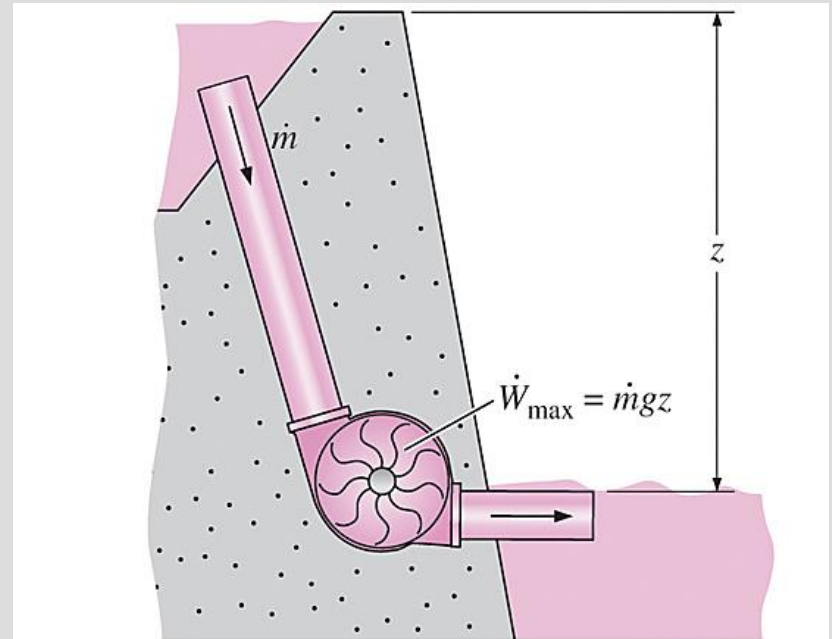
$$x_{pe} = pe = gz \quad (\text{kJ/kg})$$

Exergy of potential energy:

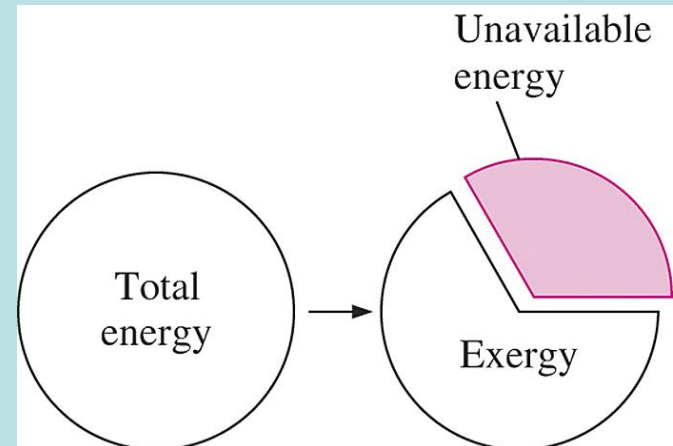
$$x_{ke} = ke = \frac{V^2}{2} \quad (\text{kJ/kg})$$

The exergies of kinetic and potential energies are equal to themselves, and they are entirely available for work.

The work potential or exergy of potential energy is equal to the potential energy itself.



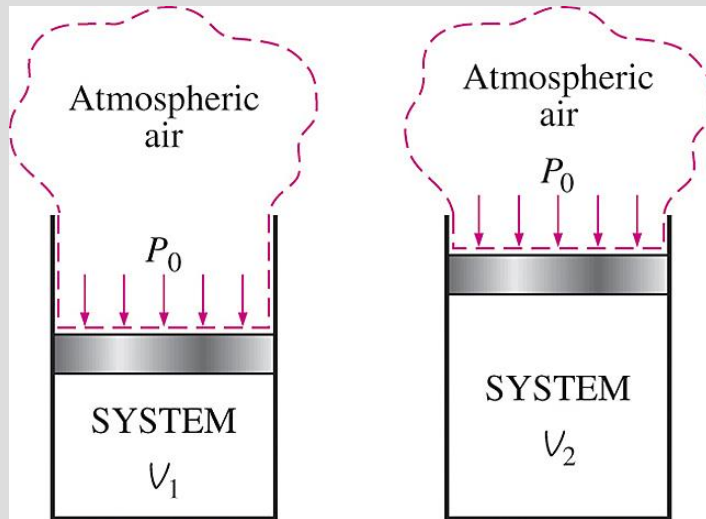
Unavailable energy is the portion of energy that cannot be converted to work by even a reversible heat engine.



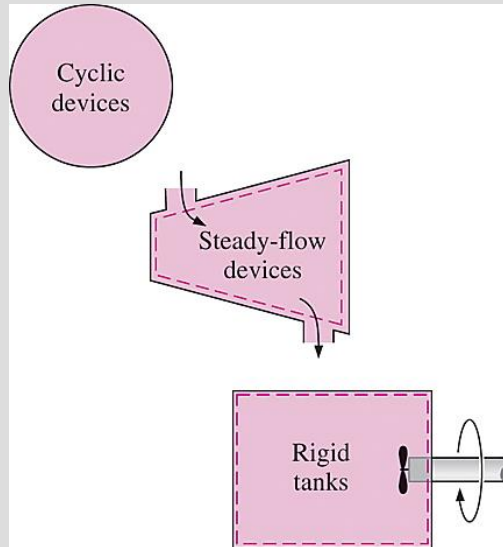
REVERSIBLE WORK AND IRREVERSIBILITY

$$W_{\text{surr}} = P_0(V_2 - V_1)$$

$$W_u = W - W_{\text{surr}} = W - P_0(V_2 - V_1)$$



As a closed system expands, some work needs to be done to push the atmospheric air out of the way (W_{surr}).



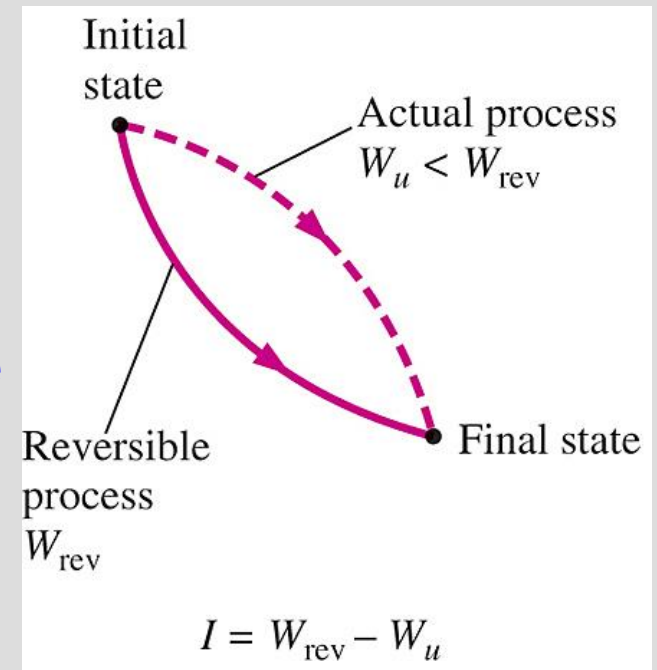
The difference between reversible work and actual useful work is the irreversibility.

For constant-volume systems, the total actual and useful works are identical ($W_u = W$).

Reversible work W_{rev} : The maximum amount of useful work that can be produced (or the minimum work that needs to be supplied) as a system undergoes a process between the specified initial and final states.

$$I = W_{\text{rev,out}} - W_{u,\text{out}}$$

$$I = W_{u,\text{in}} - W_{\text{rev,in}}$$



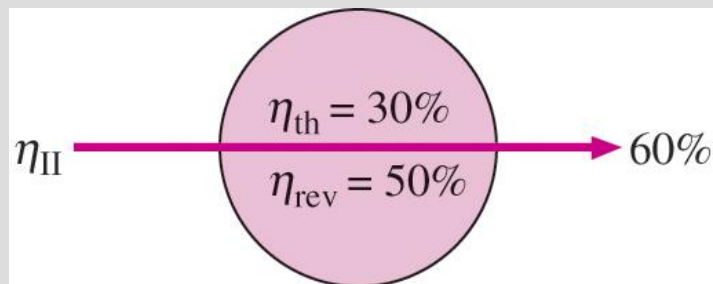
SECOND-LAW EFFICIENCY, η_{II}

$$\eta_{II} = \frac{\eta_{th}}{\eta_{th,rev}} \quad (\text{heat engines})$$

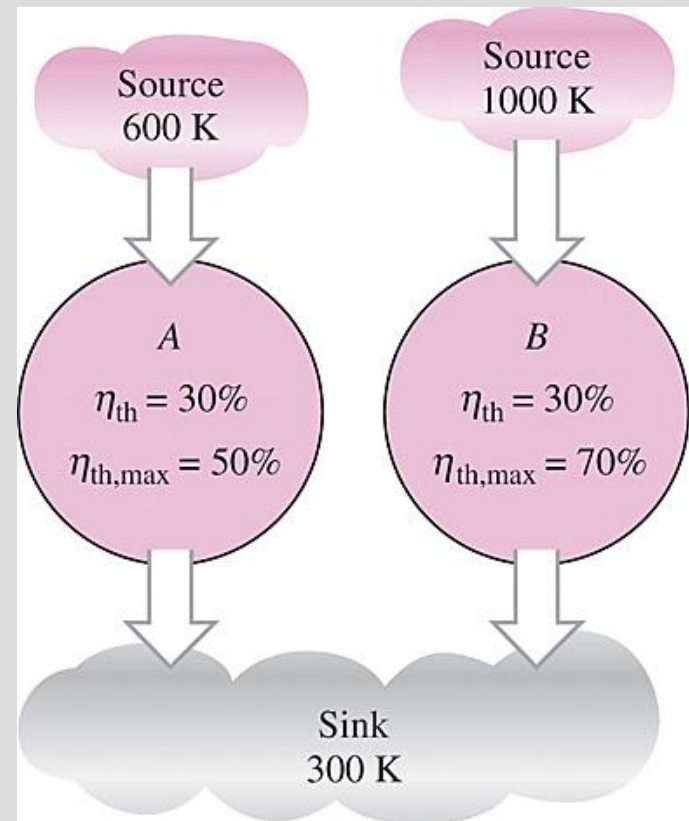
$$\eta_{II} = \frac{W_u}{W_{rev}} \quad (\text{work-producing devices})$$

$$\eta_{II} = \frac{W_{rev}}{W_u} \quad (\text{work-consuming devices})$$

$$\eta_{II} = \frac{COP}{COP_{rev}} \quad (\text{refrigerators and heat pumps})$$



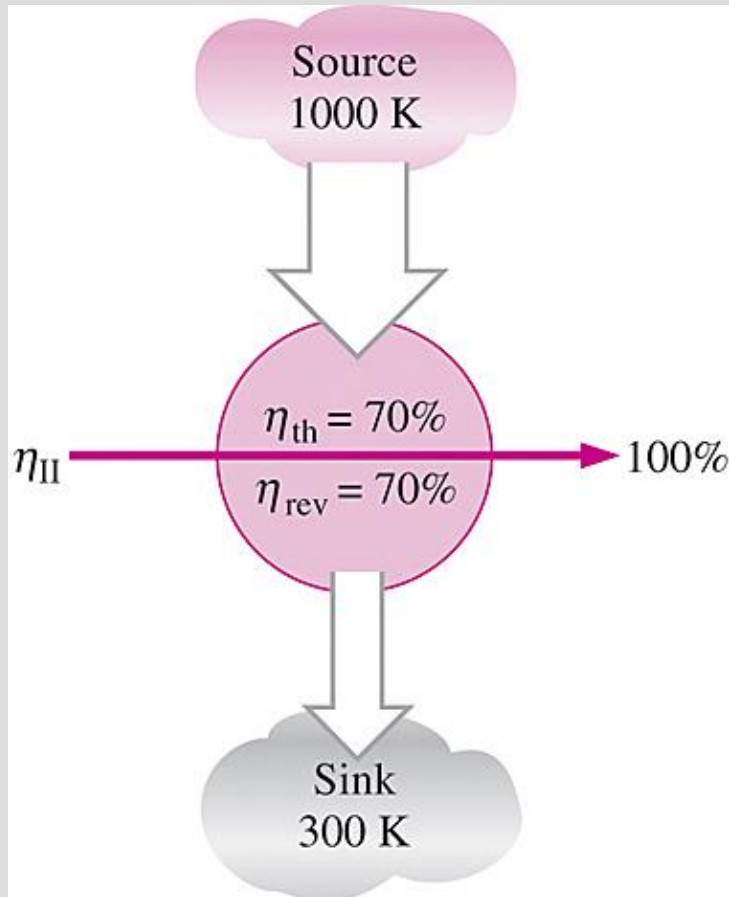
Second-law efficiency is a measure of the performance of a device relative to its performance under reversible conditions.



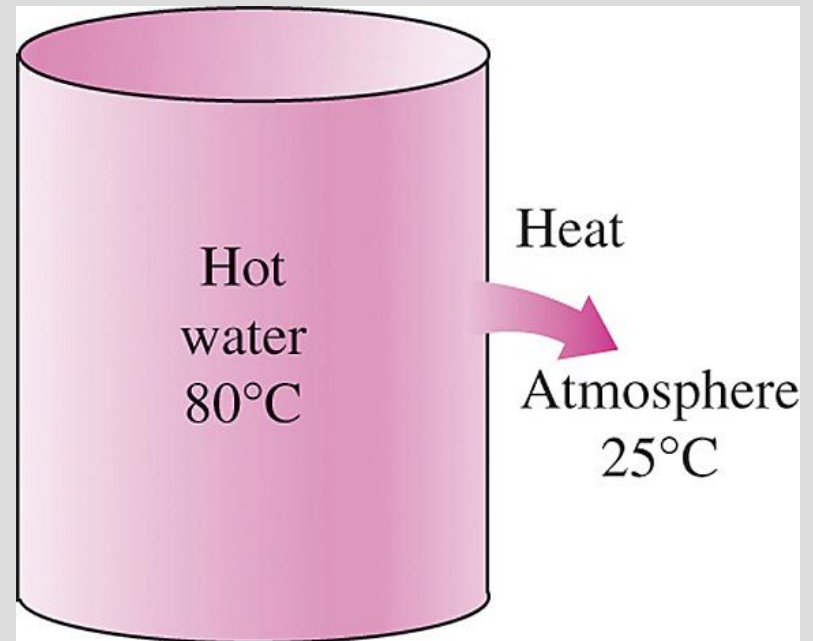
Two heat engines that have the same thermal efficiency, but different maximum thermal efficiencies.

$$\eta_{II} = \frac{\text{Exergy recovered}}{\text{Exergy supplied}} = 1 - \frac{\text{Exergy destroyed}}{\text{Exergy supplied}}$$

General definition of exergy efficiency



Second-law efficiency of all reversible devices is 100%.



The second-law efficiency of naturally occurring processes is zero if none of the work potential is recovered.

EXERGY CHANGE OF A SYSTEM

Exergy of a Fixed Mass: Nonflow (or Closed System) Exergy

$$\underbrace{\delta E_{\text{in}} - \delta E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{dE_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc., energies}}$$

$$- \delta Q - \delta W = dU$$

$$\delta W = P dV = (P - P_0) dV + P_0 dV = \delta W_{b,\text{useful}} + P_0 dV$$

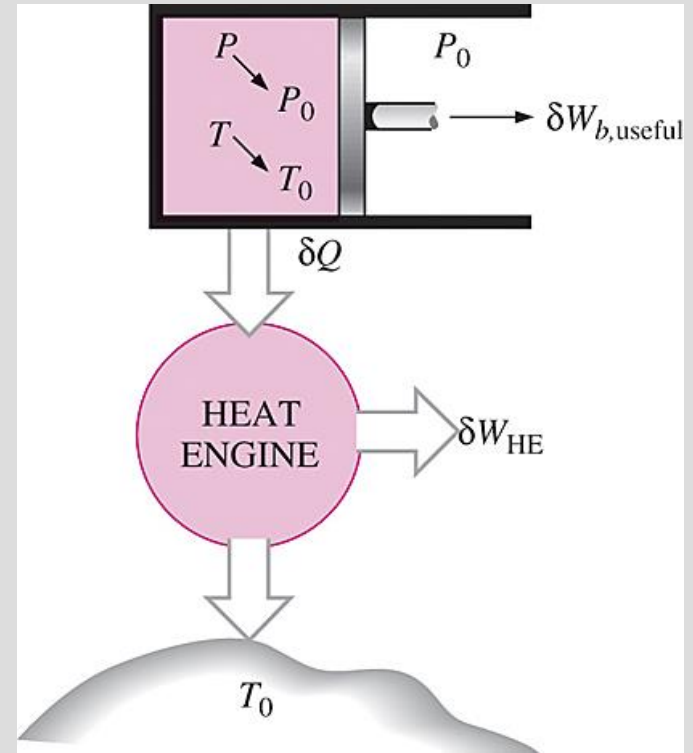
$$\delta W_{\text{HE}} = \left(1 - \frac{T_0}{T}\right) \delta Q = \delta Q - \frac{T_0}{T} \delta Q = \delta Q - (-T_0 dS) \rightarrow$$

$$\delta Q = \delta W_{\text{HE}} - T_0 dS$$

$$\delta W_{\text{total useful}} = \delta W_{\text{HE}} + \delta W_{b,\text{useful}} = -dU - P_0 dV + T_0 dS$$

$$X = (U - U_0) + P_0(V - V_0) - T_0(S - S_0) + m \frac{V^2}{2} + mgz$$

Exergy of a closed system



The exergy of a specified mass at a specified state is the useful work that can be produced as the mass undergoes a reversible process to the state of the environment.

$$\phi = (u - u_0) + P_0(v - v_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

$$= (e - e_0) + P_0(v - v_0) - T_0(s - s_0)$$

Closed system
exergy per unit
mass

$$\Delta X = X_2 - X_1 = m(\phi_2 - \phi_1) = (E_2 - E_1) + P_0(V_2 - V_1) - T_0(S_2 - S_1)$$

$$= (U_2 - U_1) + P_0(V_2 - V_1) - T_0(S_2 - S_1) + m \frac{V_2^2 - V_1^2}{2} + mg(z_2 - z_1)$$

Exergy
change of
a closed
system

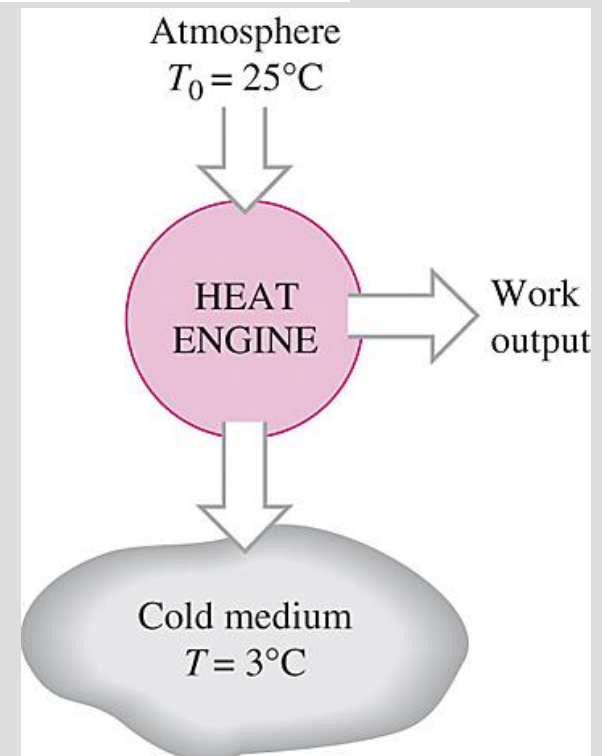
$$\Delta \phi = \phi_2 - \phi_1 = (u_2 - u_1) + P_0(v_2 - v_1) - T_0(s_2 - s_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

$$= (e_2 - e_1) + P_0(v_2 - v_1) - T_0(s_2 - s_1)$$

When the properties of a system are not uniform, the exergy of the system is

$$X_{\text{system}} = \int \phi \delta m = \int_V \phi \rho dV$$

The exergy of a cold medium is also a *positive* quantity since work can be produced by transferring heat to it.



Exergy of a Flow Stream: Flow (or Stream) Exergy

$$x_{\text{flowing fluid}} = x_{\text{nonflowing fluid}} + x_{\text{flow}}$$

$$= (u - u_0) + P_0(\nu - \nu_0) - T_0(s - s_0) + \frac{V^2}{2} + gz + (P - P_0)\nu$$

$$= (u + P\nu) - (u_0 + P_0\nu_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

$$= (h - h_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

Exergy of flow energy

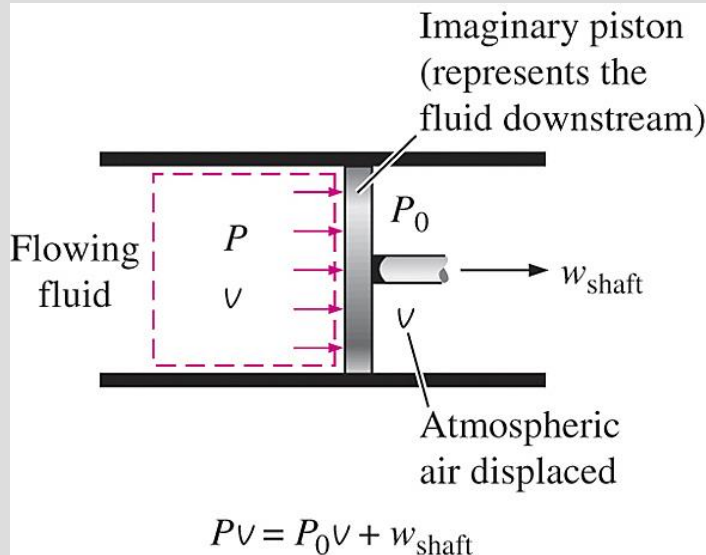
$$x_{\text{flow}} = P\nu - P_0\nu = (P - P_0)\nu$$

Flow
exergy

$$\psi = (h - h_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

Exergy change of flow

$$\Delta\psi = \psi_2 - \psi_1 = (h_2 - h_1) + T_0(s_2 - s_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$



The *exergy* associated with *flow energy* is the useful work that would be delivered by an imaginary piston in the flow section.

Energy:

$$e = u + \frac{V^2}{2} + gz$$

Exergy:

$$\phi = (u - u_0) + P_0(v - v_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$



(a) A fixed mass (nonflowing)

Energy:

$$\theta = h + \frac{V^2}{2} + gz$$

Exergy:

$$\psi = (h - h_0) + T_0(s - s_0) + \frac{V^2}{2} + gz$$



(b) A fluid stream (flowing)

The *energy* and *exergy* contents of
(a) a fixed mass
(b) a fluid stream.

EXERGY TRANSFER BY HEAT, WORK, AND MASS

Exergy by Heat Transfer, Q

$$X_{\text{heat}} = \left(1 - \frac{T_0}{T}\right) Q$$

Exergy transfer by heat

$$X_{\text{heat}} = \int \left(1 - \frac{T_0}{T}\right) \delta Q$$

When temperature is not constant

HEAT SOURCE

Temperature: T

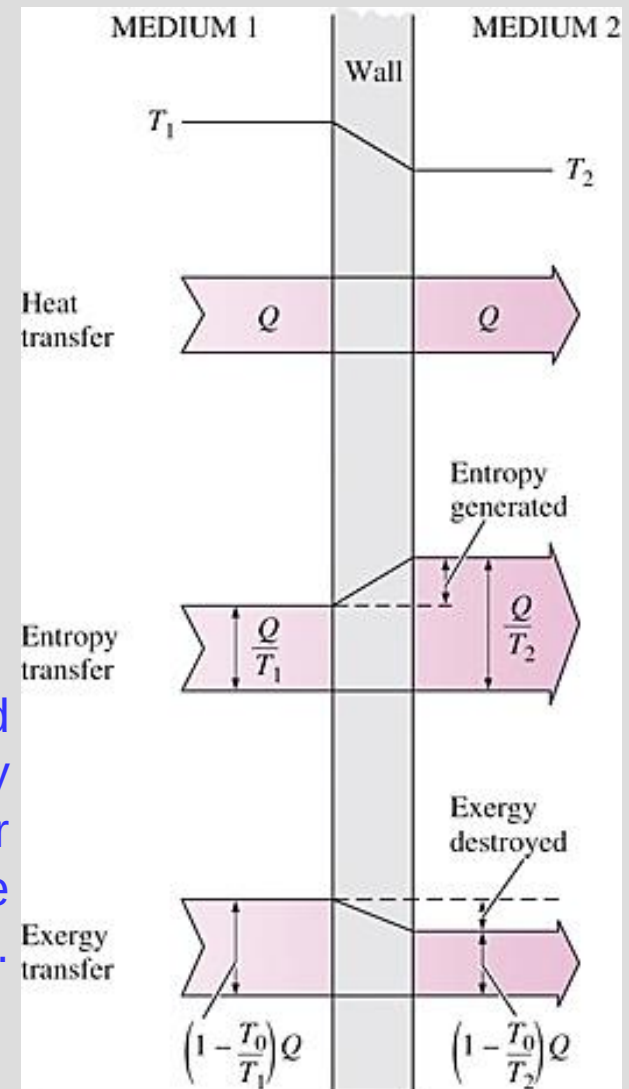
Energy transferred: E

Exergy = $\left(1 - \frac{T_0}{T}\right) E$

T_0

The transfer and destruction of exergy during a heat transfer process through a finite temperature difference.

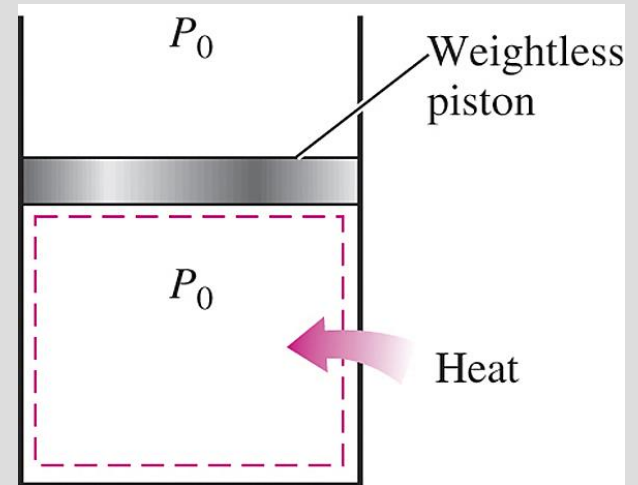
The Carnot efficiency $\eta_c = 1 - T_0/T$ represents the fraction of the energy transferred from a heat source at temperature T that can be converted to work in an environment at temperature T_0 .



Exergy Transfer by Work, W

$$X_{\text{work}} = \begin{cases} W - W_{\text{surr}} & \text{(for boundary work)} \\ W & \text{(for other forms of work)} \end{cases}$$

$$W_{\text{surr}} = P_0(V_2 - V_1)$$

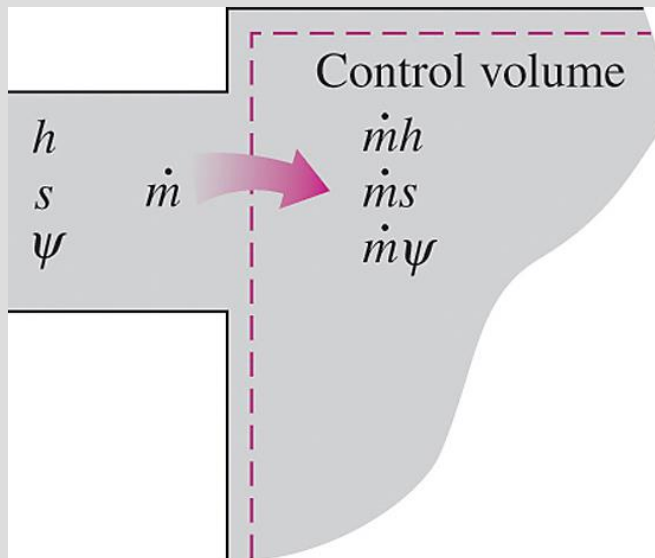


There is no useful work transfer associated with boundary work when the pressure of the system is maintained constant at atmospheric pressure.

Exergy Transfer by Mass, m

$$X_{\text{mass}} = m\psi \quad \psi = (h - h_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$$

$$\dot{X}_{\text{mass}} = \int_{A_c} \psi \rho V_n dA_c \quad X_{\text{mass}} = \int \psi \delta m = \int_{\Delta t} \dot{X}_{\text{mass}} dt$$



Mass contains energy, entropy, and exergy, and thus mass flow into or out of a system is accompanied by energy, entropy, and exergy transfer.

THE DECREASE OF EXERGY PRINCIPLE AND EXERGY DESTRUCTION

Energy balance: $E_{in}^0 - E_{out}^0 = \Delta E_{system} \rightarrow 0 = E_2 - E_1$

Entropy balance: $S_{in}^0 - S_{out}^0 + S_{gen} = \Delta S_{system} \rightarrow S_{gen} = S_2 - S_1$

Multiplying the second relation by T_0 and subtracting it from the first one gives

$$-T_0 S_{gen} = E_2 - E_1 - T_0(S_2 - S_1) \quad (8-29)$$

From Eq. 8-17 we have

$$\begin{aligned} X_2 - X_1 &= (E_2 - E_1) + P_0(V_2 - V_1)^0 - T_0(S_2 - S_1) \\ &= (E_2 - E_1) - T_0(S_2 - S_1) \end{aligned} \quad (8-30)$$

since $V_2 = V_1$ for an isolated system (it cannot involve any moving boundary and thus any boundary work). Combining Eqs. 8-29 and 8-30 gives

$$-T_0 S_{gen} = X_2 - X_1 \leq 0 \quad (8-31)$$

since T_0 is the thermodynamic temperature of the environment and thus a positive quantity, $S_{gen} \geq 0$, and thus $T_0 S_{gen} \geq 0$. Then we conclude that

$$\Delta X_{isolated} = (X_2 - X_1)_{isolated} \leq 0 \quad (8-32)$$

No heat, work
or mass transfer

Isolated system

$$\Delta X_{isolated} \leq 0$$

$$(\text{or } X_{\text{destroyed}} \geq 0)$$

The isolated system considered in the development of the decrease of exergy principle.

*The exergy of an isolated system during a process always decreases or, in the limiting case of a reversible process, remains constant. In other words, it never increases and exergy is destroyed during an actual process. This is known as the **decrease of exergy principle**.*

Exergy Destruction

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} \geq 0$$

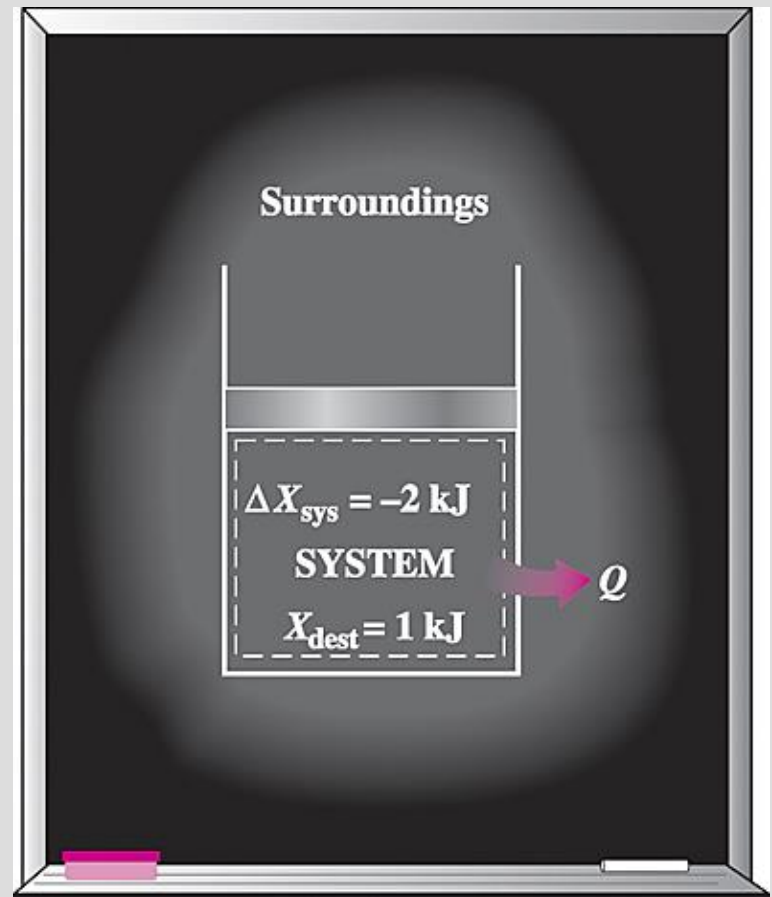
$$X_{\text{destroyed}} \begin{cases} > 0 & \text{Irreversible process} \\ = 0 & \text{Reversible process} \\ < 0 & \text{Impossible process} \end{cases}$$

Exergy destroyed is a *positive quantity* for any actual process and becomes *zero* for a reversible process.

Exergy destroyed represents the lost work potential and is also called the *irreversibility* or *lost work*.

Can the exergy change of a system during a process be negative?

Consider heat transfer from a system to its surroundings. How do you compare exergy changes of the system and the surroundings?



The exergy change of a system can be negative, but the exergy destruction cannot.

EXERGY BALANCE: CLOSED SYSTEMS

$$\left(\begin{array}{c} \text{Total} \\ \text{exergy} \\ \text{entering} \end{array} \right) - \left(\begin{array}{c} \text{Total} \\ \text{exergy} \\ \text{leaving} \end{array} \right) - \left(\begin{array}{c} \text{Total} \\ \text{exergy} \\ \text{destroyed} \end{array} \right) = \left(\begin{array}{c} \text{Change in the} \\ \text{total exergy} \\ \text{of the system} \end{array} \right)$$

General:

$$\underbrace{X_{\text{in}} - X_{\text{out}}}_{\substack{\text{Net exergy transfer} \\ \text{by heat, work, and mass}}} - \underbrace{X_{\text{destroyed}}}_{\substack{\text{Exergy} \\ \text{destruction}}} = \underbrace{\Delta X_{\text{system}}}_{\substack{\text{Change} \\ \text{in exergy}}} \quad (\text{kJ})$$

General, rate form:

$$\underbrace{\dot{X}_{\text{in}} - \dot{X}_{\text{out}}}_{\substack{\text{Rate of net exergy transfer} \\ \text{by heat, work, and mass}}} - \underbrace{\dot{X}_{\text{destroyed}}}_{\substack{\text{Rate of exergy} \\ \text{destruction}}} = \underbrace{dX_{\text{system}}/dt}_{\substack{\text{Rate of change} \\ \text{in exergy}}} \quad (\text{kW})$$

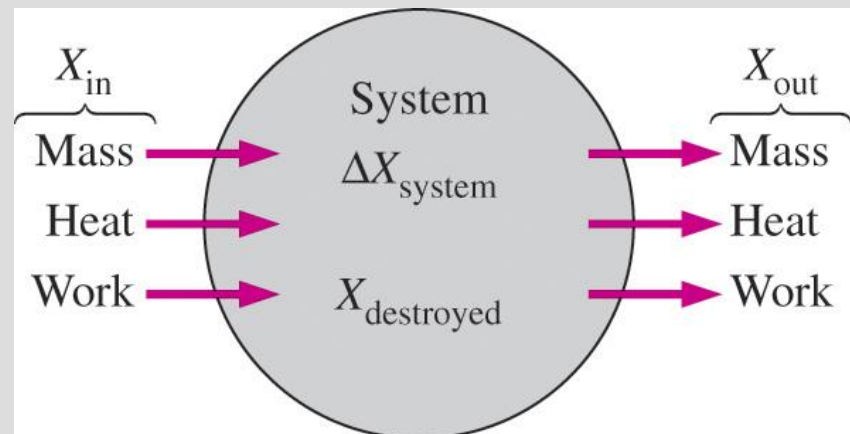
$$\dot{X}_{\text{heat}} = (1 - T_0/T)\dot{Q}, \quad \dot{X}_{\text{work}} = \dot{W}_{\text{useful}}, \quad \text{and} \quad \dot{X}_{\text{mass}} = \dot{m}\psi$$

General, unit-mass basis:

$$(x_{\text{in}} - x_{\text{out}}) - x_{\text{destroyed}} = \Delta x_{\text{system}} \quad (\text{kJ/kg})$$

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} \quad \text{or} \quad \dot{X}_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}}$$

Mechanisms
of exergy
transfer.



The exergy change of a system during a process is equal to the difference between the net exergy transfer through the system boundary and the exergy destroyed within the system boundaries as a result of irreversibilities.

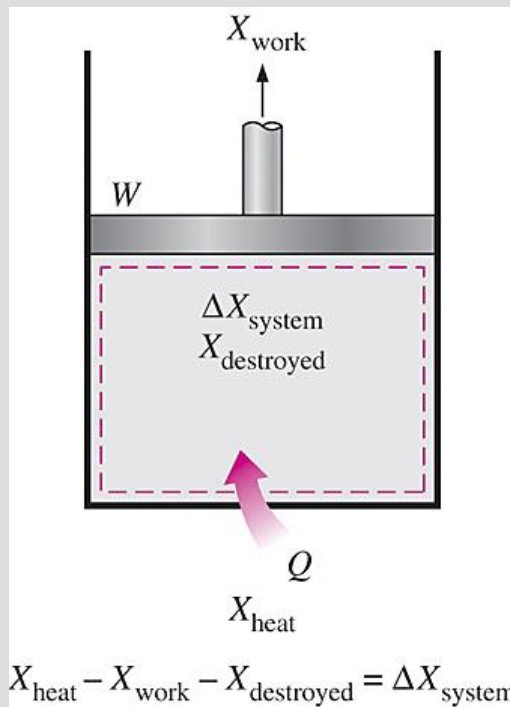
Closed system:
$$X_{\text{heat}} - X_{\text{work}} - X_{\text{destroyed}} = \Delta X_{\text{system}}$$

Closed system:
$$\sum \left(1 - \frac{T_0}{T_k}\right) Q_k - [W - P_0(V_2 - V_1)] - T_0 S_{\text{gen}} = X_2 - X_1$$

Rate form:
$$\sum \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_k - \left(\dot{W} - P_0 \frac{dV_{\text{system}}}{dt}\right) - T_0 \dot{S}_{\text{gen}} = \frac{dX_{\text{system}}}{dt}$$

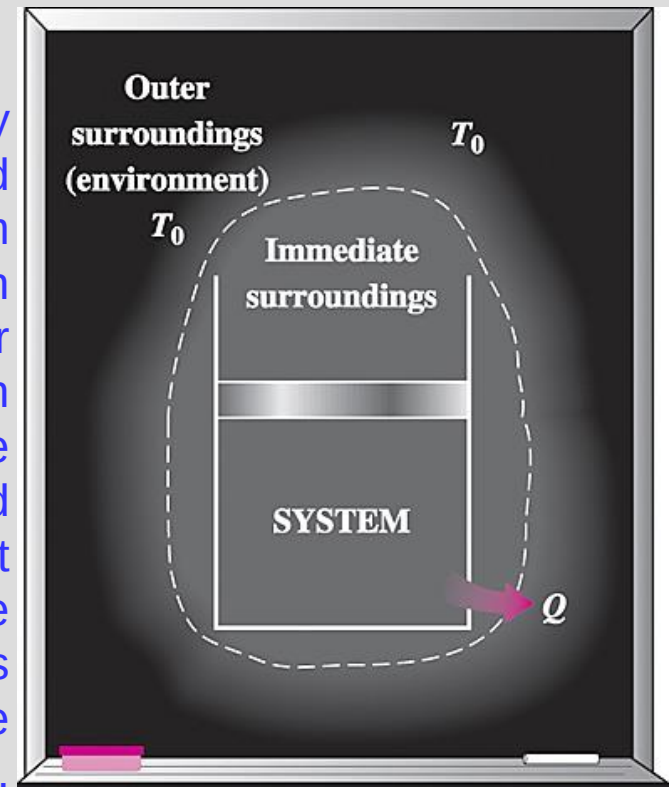
Q_k is the heat transfer through the boundary at temperature T_k at location k .

The heat transfer to a system and work done by the system are taken to be positive quantities.



Exergy balance for a closed system when heat transfer is to the system and the work is from the system.

Exergy destroyed outside system boundaries can be accounted for by writing an exergy balance on the extended system that includes the system and its immediate surroundings.

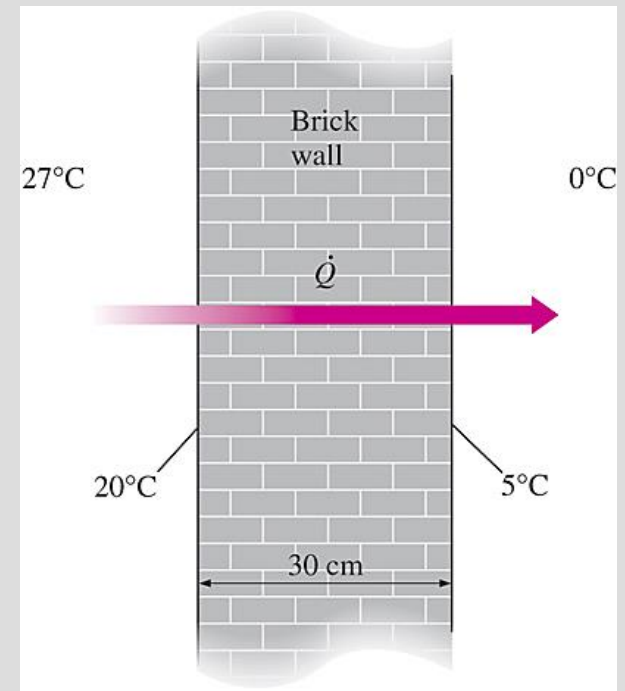


EXAMPLES

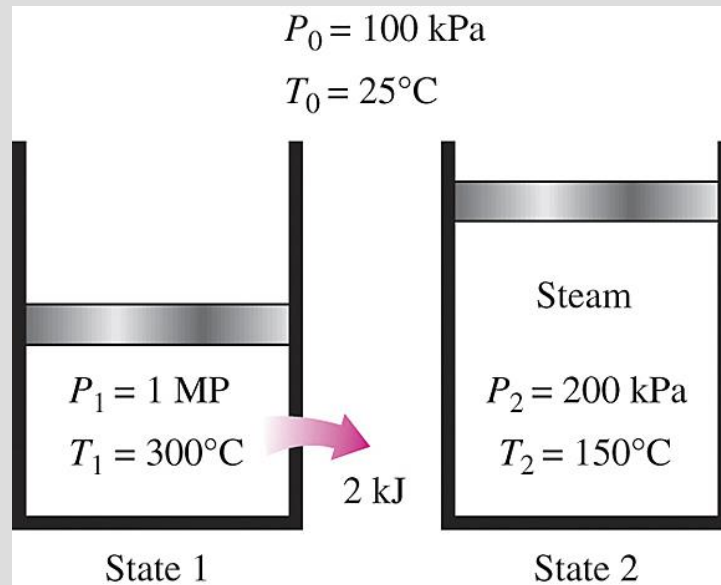
Exergy balance for heat conduction

$$\underbrace{\dot{X}_{in} - \dot{X}_{out}}_{\text{Rate of net exergy transfer by heat, work, and mass}} - \underbrace{\dot{X}_{destroyed}}_{\text{Rate of exergy destruction}} = \underbrace{dX_{system}/dt}_{\text{Rate of change in exergy}} \xrightarrow{0 \text{ (steady)}} = 0$$

$$\dot{Q} \left(1 - \frac{T_0}{T} \right)_{in} - \dot{Q} \left(1 - \frac{T_0}{T} \right)_{out} - \dot{X}_{destroyed} = 0$$



Exergy balance for expansion of steam



The exergy balance applied on the *extended system* (system + immediate surroundings) whose boundary is at the environment temperature of T_0 gives

$$\underbrace{X_{in} - X_{out}}_{\text{Net exergy transfer by heat, work, and mass}} - \underbrace{X_{destroyed}}_{\text{Exergy destruction}} = \underbrace{\Delta X_{system}}_{\text{Change in exergy}}$$

$$-X_{work,out} - X_{heat,out} \xrightarrow{0} -X_{destroyed} = X_2 - X_1$$

$$X_{destroyed} = X_1 - X_2 - W_{u,out}$$

Exergy balance for an air tank

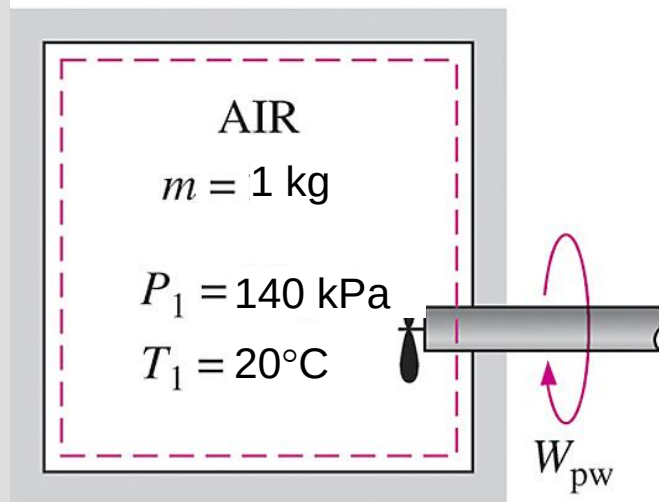
$$\underbrace{X_{\text{in}} - X_{\text{out}}}_{\text{Net exergy transfer by heat, work, and mass}} - \underbrace{X_{\text{destroyed}}}_{\text{Exergy destruction}} \overset{0 \text{ (reversible)}}{\rightarrow} = \underbrace{\Delta X_{\text{system}}}_{\text{Change in exergy}}$$

$$W_{\text{rev,in}} = X_2 - X_1$$

$$= (E_2 - E_1) + P_0(V_2 - V_1) \overset{0}{\rightarrow} - T_0(S_2 - S_1)$$

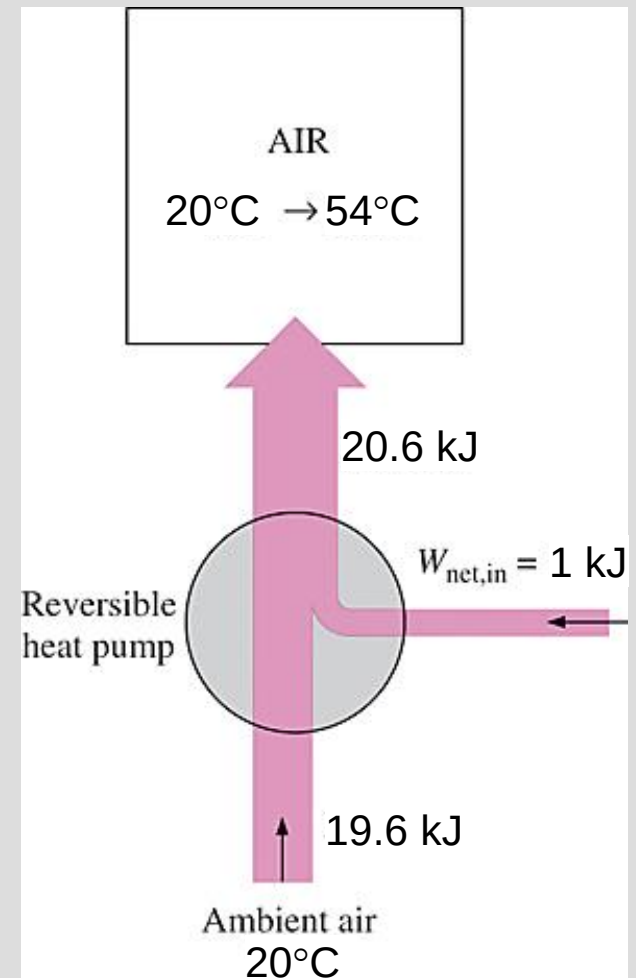
$$= (U_2 - U_1) - T_0(S_2 - S_1) = 1 \text{ kJ}$$

$$T_0 = 20^\circ\text{C}$$



$$W_{\text{pw,in}} = \Delta U = 20.6 \text{ kJ}$$

$$W_{\text{rev,in}} = 1 \text{ kJ}$$



The same effect on the insulated tank system can be accomplished by a reversible heat pump that consumes only 1 kJ of work.

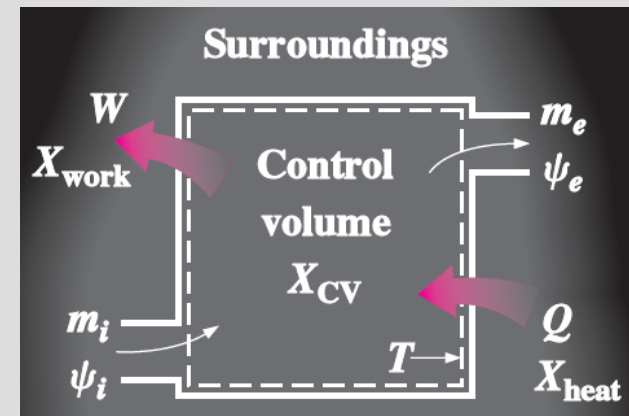
EXERGY BALANCE: CONTROL VOLUMES

$$X_{\text{heat}} - X_{\text{work}} + X_{\text{mass,in}} - X_{\text{mass,out}} - X_{\text{destroyed}} = (X_2 - X_1)_{\text{CV}}$$

$$\sum \left(1 - \frac{T_0}{T_k}\right) Q_k - [W - P_0(V_2 - V_1)] + \sum_{\text{in}} m\psi - \sum_{\text{out}} m\psi - X_{\text{destroyed}} = (X_2 - X_1)_{\text{CV}}$$

$$\sum \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_k - \left(\dot{W} - P_0 \frac{dV_{\text{CV}}}{dt}\right) + \sum_{\text{in}} \dot{m}\psi - \sum_{\text{out}} \dot{m}\psi - \dot{X}_{\text{destroyed}} = \frac{dX_{\text{CV}}}{dt}$$

The rate of exergy change within the control volume during a process is equal to the rate of net exergy transfer through the control volume boundary by heat, work, and mass flow minus the rate of exergy destruction within the boundaries of the control volume.



Exergy is transferred into or out of a control volume by mass as well as heat and work transfer.

Exergy Balance for Steady-Flow Systems

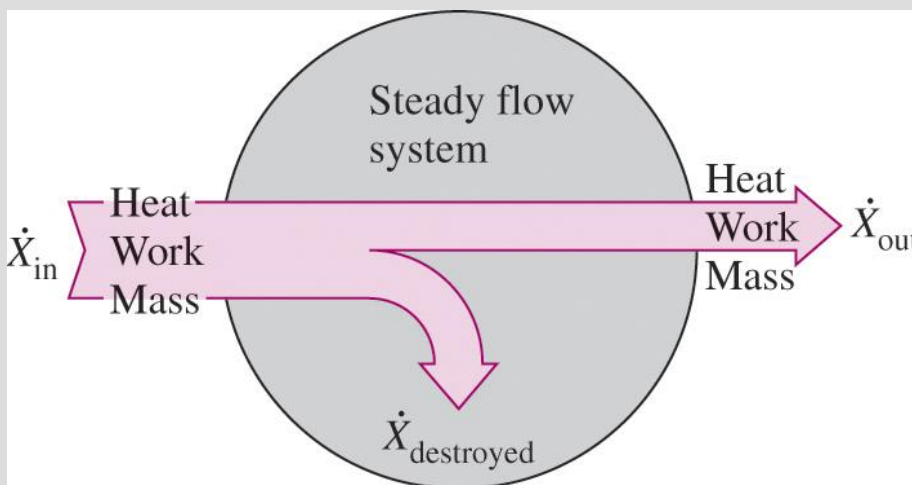
Most control volumes encountered in practice such as turbines, compressors, nozzles, diffusers, heat exchangers, pipes, and ducts operate steadily, and thus they experience no changes in their mass, energy, entropy, and exergy contents as well as their volumes. Therefore, $dV_{CV}/dt = 0$ and $dX_{CV}/dt = 0$ for such systems.

$$\text{Steady-flow:} \quad \sum \left(1 - \frac{T_0}{T_k} \right) \dot{Q}_k - \dot{W} + \sum_{\text{in}} \dot{m}\psi - \sum_{\text{out}} \dot{m}\psi - \dot{X}_{\text{destroyed}} = 0$$

$$\text{Single-stream:} \quad \sum \left(1 - \frac{T_0}{T_k} \right) \dot{Q}_k - \dot{W} + \dot{m}(\psi_1 - \psi_2) - \dot{X}_{\text{destroyed}} = 0$$

$$\psi_1 - \psi_2 = (h_1 - h_2) - T_0(s_1 - s_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2)$$

$$\text{Per-unit mass:} \quad \sum \left(1 - \frac{T_0}{T_k} \right) q_k - w + (\psi_1 - \psi_2) - x_{\text{destroyed}} = 0$$



The exergy transfer to a steady-flow system is equal to the exergy transfer from it plus the exergy destruction within the system.

Reversible Work, W_{rev}

The exergy balance relations presented above can be used to determine the reversible work W_{rev} by setting the exergy destroyed equal to zero. The work W in that case becomes the reversible work.

$$\text{General:} \quad W = W_{\text{rev}} \quad \text{when } X_{\text{destroyed}} = 0$$

$$\text{Single stream:} \quad \dot{W}_{\text{rev}} = \dot{m}(\psi_1 - \psi_2) + \sum \left(1 - \frac{T_0}{T_k} \right) \dot{Q}_k \quad (\text{kW})$$

$$\text{Adiabatic, single stream:} \quad \dot{W}_{\text{rev}} = \dot{m}(\psi_1 - \psi_2)$$

The exergy destroyed is zero only for a reversible process, and reversible work represents the maximum work output for work-producing devices such as turbines and the minimum work input for work-consuming devices such as compressors.

Second-Law Efficiency of Steady-Flow Devices, η_{II}

The *second-law efficiency* of various steady-flow devices can be determined from its general definition, $\eta_{II} = (\text{Exergy recovered})/(\text{Exergy supplied})$. When the changes in kinetic and potential energies are negligible and the devices are **adiabatic**:

$$\eta_{II,\text{turb}} = \frac{w}{w_{\text{rev}}} = \frac{h_1 - h_2}{\psi_1 - \psi_2} \quad \text{or} \quad \eta_{II,\text{turb}} = 1 - \frac{T_0 s_{\text{gen}}}{\psi_1 - \psi_2}$$

Turbine

$$s_{\text{gen}} = s_2 - s_1$$

$$\eta_{II,\text{comp}} = \frac{w_{\text{rev,in}}}{w_{\text{in}}} = \frac{\psi_2 - \psi_1}{h_2 - h_1} \quad \text{or} \quad \eta_{II,\text{comp}} = 1 - \frac{T_0 s_{\text{gen}}}{h_2 - h_1}$$

Compressor

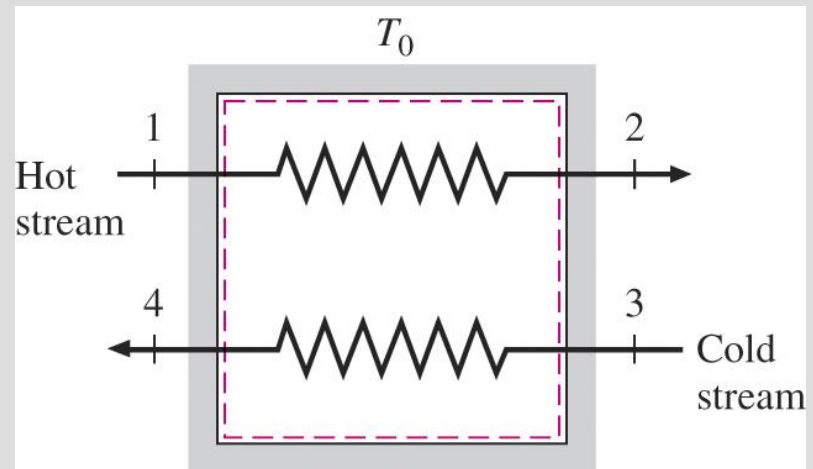
$$\eta_{II,\text{HX}} = \frac{\dot{m}_{\text{cold}}(\psi_4 - \psi_3)}{\dot{m}_{\text{hot}}(\psi_1 - \psi_2)} \quad \text{Heat exchanger}$$

$$\eta_{II,\text{HX}} = 1 - \frac{T_0 \dot{S}_{\text{gen}}}{\dot{m}_{\text{hot}}(\psi_1 - \psi_2)}$$

$$\dot{S}_{\text{gen}} = \dot{m}_{\text{hot}}(s_2 - s_1) + \dot{m}_{\text{cold}}(s_4 - s_3)$$

$$\eta_{II,\text{mix}} = \frac{\dot{m}_3 \psi_3}{\dot{m}_1 \psi_1 + \dot{m}_2 \psi_2} \quad \text{Mixing chamber}$$

$$\eta_{II,\text{mix}} = 1 - \frac{T_0 \dot{S}_{\text{gen}}}{\dot{m}_1 \psi_1 + \dot{m}_2 \psi_2} \quad \dot{S}_{\text{gen}} = \dot{m}_3 s_3 - \dot{m}_2 s_2 - \dot{m}_1 s_1$$



A heat exchanger with two unmixed fluid streams.

EXAMPLES

Exergy analysis of a steam turbine

$$\underbrace{\dot{X}_{in} - \dot{X}_{out}}_{\text{Rate of net exergy transfer by heat, work, and mass}} - \underbrace{\dot{X}_{destroyed}}_{\text{Rate of exergy destruction}} \xrightarrow{0 \text{ (reversible)}} = \underbrace{dX_{system}/dt}_{\text{Rate of change in exergy}} \xrightarrow{0 \text{ (steady)}} = 0$$

$$\dot{X}_{in} = \dot{X}_{out}$$

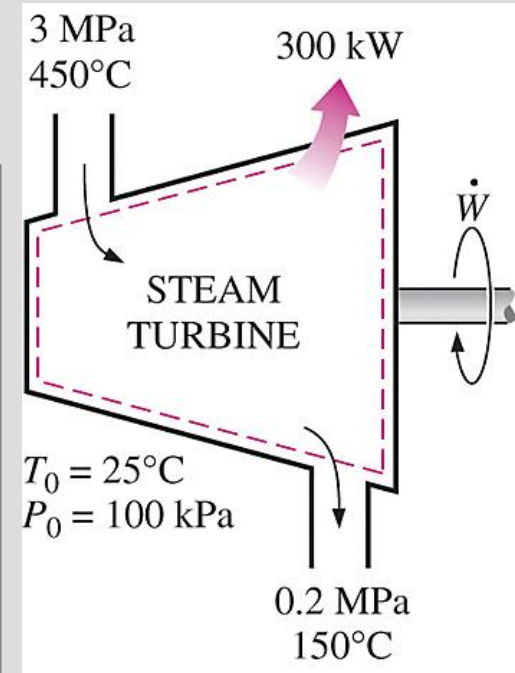
$$\dot{m}\psi_1 = \dot{W}_{rev,out} + \dot{X}_{heat} \xrightarrow{0} + \dot{m}\psi_2$$

$$\dot{W}_{rev,out} = \dot{m}(\psi_1 - \psi_2)$$

$$\dot{X}_{destroyed} = \dot{W}_{rev,out} - \dot{W}_{out}$$

$$= \dot{m}[(h_1 - h_2) - T_0(s_1 - s_2) - \Delta ke \xrightarrow{0} - \Delta pe \xrightarrow{0}]$$

$$\eta_{II} = \frac{\dot{W}_{out}}{\dot{W}_{rev,out}}$$



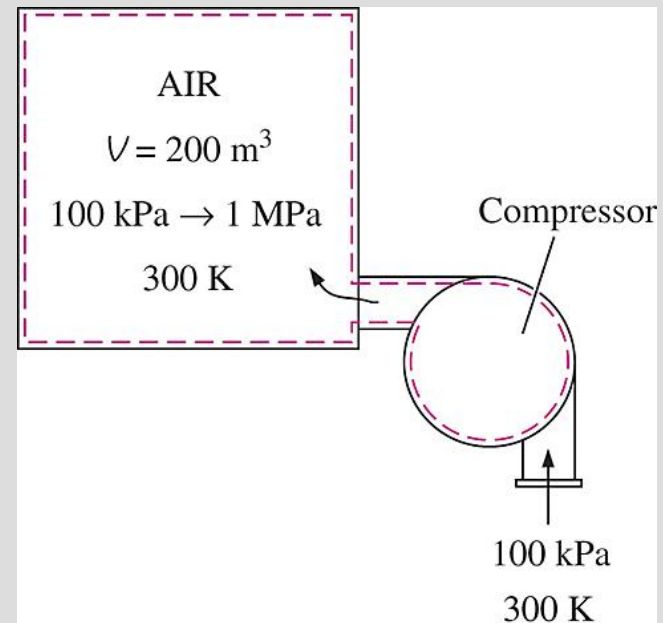
Exergy balance for a charging process

$$\underbrace{X_{in} - X_{out}}_{\text{Net exergy transfer by heat, work, and mass}} - \underbrace{X_{destroyed}}_{\text{Exergy destruction}} \xrightarrow{0 \text{ (reversible)}} = \underbrace{\Delta X_{system}}_{\text{Change in exergy}}$$

$$X_{in} - X_{out} = X_2 - X_1$$

$$W_{rev,in} + m_1\psi_1 \xrightarrow{0} = m_2\phi_2 - m_1\phi_1 \xrightarrow{0}$$

$$W_{rev,in} = m_2\phi_2$$



Summary

- Exergy: Work potential of energy
 - ✓ Exergy (work potential) associated with kinetic and potential energy
- Reversible work and irreversibility
- Second-law efficiency
- Exergy change of a system
 - ✓ Exergy of a fixed mass: Nonflow (or closed system) exergy
 - ✓ Exergy of a flow stream: Flow (or stream) exergy
- Exergy transfer by heat, work, and mass
- The decrease of exergy principle and exergy destruction
- Exergy balance: Closed systems
- Exergy balance: Control volumes
 - ✓ Exergy balance for steady-flow systems
 - ✓ Reversible work
 - ✓ Second-law efficiency of steady-flow devices